

## Dart Programming Language

### \* String Buffer :-

#### Syntax :

```
void main() {  
  ✓ StringBuffer buffer = StringBuffer();  
  ✓ or  
  var buffer = StringBuffer();  
  print(buffer);  
}
```

```
void main() {  
  String name = "Coder";  
  ✓ name += " ";  
  name += "Club";  
  ✓ name += " ";  
  name += "Bangla";  
  print(name);  
}
```

'Coder Club Bangla'

```
void main() {  
  StringBuffer buffer = StringBuffer();
```

### \* write()

```
{  
  buffer.write("Coder");  
  buffer.write(" ");  
  buffer.write("Club");  
  buffer.write(" ");  
  buffer.write("Bangla");  
  print(buffer);  
}
```

Coder Club Bangla

### \* writeLn()

```
void main() {  
  StringBuffer buffer = StringBuffer();  
  buffer.writeLn("Apple");  
}
```

```

buffer. writeLn ("Banana");
buffer. writeLn ("Mango");
print (buffer);
}
// Apple
// Banana
// Mango

```

\* Write All

```

void main() {
    List<String> fruits = ["apple", "Banana", "Mango"];
    StringBuffer buffer = StringBuffer();
    ✓ buffer. writeAll (fruits);
    print (buffer);
}
// appleBananaMango

```

```

buffer. writeAll (fruits, " - ");

```

apple - Banana - Mango

Loop with String Buffer :-

```

void main() {
    StringBuffer buffer = StringBuffer();
    for (int i = 1; i <= 5; i++) {
        buffer. write ("5 * %i ");
    }
    print (buffer); // 1 2 3 4 5
}

```

```

void main()
var buffer = StringBuffer();
for (int i = 1; i <= 10; i++) {
    buffer. writeLn ("5 * %i = %i");
}
print (buffer);
// 5 * 1 = 5
// 5 * 2 = 10
// ...
// 5 * 10 = 50 //

```

length

```

void main() {
    var buffer = StringBuffer();
    buffer. write ("Hello World");
    print (buffer. length);
}
// 10

```

\* isEmpty , isNotEmpty

```
void main() {  
    var buffer = StringBuffer();  
    print(buffer.isEmpty);  
    buffer.write("Dart"); //true ✓  
    print(buffer.isNotEmpty);  
    //true  
}
```

\* clear()

```
void main() {  
    StringBuffer buffer = StringBuffer();  
    buffer.write("Hello");  
    buffer.clear();  
    print(buffer);  
}
```

\* toString()

```
void main() {  
    StringBuffer buffer = StringBuffer();  
    buffer.write("Coder Club Bangla");  
    String result = buffer.toString();  
    print(result);  
} // Coder Club Bangla
```

\* HTML

```
void main() {  
    StringBuffer buffer = StringBuffer();  
    buffer.writeln("<html>");  
    buffer.writeln("<body>");  
    buffer.writeln("<h1> Hello Dart </h1>");  
    buffer.writeln("</body>");  
    buffer.writeln("</html>");  
    print(buffer);  
}
```

```
// <html>  
// <body>  
// <h1> Hello Dart </h1>  
// </body>  
// </html>
```

## String Concatenation Vs String Buffer Performance

```
void main() {  
    String result = "";  
    for (int i = 0; i < 10000; i++) {  
        result += ' ' + i;  
    }  
    print(result);  
}
```

0  
↓  
01  
↓  
012  
↓  
0123  
↓

result = result + ' ' + i;  
✓ 0 = "" + 0  
✓ 01 = 0 + 1  
✓ 012 = 01 + 2  
✓ 0123 = 012 + 3  
  
+=

```
void main() {  
    StringBuffer buffer = StringBuffer();  
    for (int i = 0; i < 10000; i++) {  
        buffer.write(i);  
    }  
    print(buffer);  
}
```

buffer  
→ 0  
→ 1  
→ 2  
→ 3  
→ 4  
→  
→  
→  
→ 9999

0 1 2 3 4 . . .

### Homework

| Student      | Report |
|--------------|--------|
| Name :       | Rahim  |
| Age :        | 20     |
| Department : | CSE    |
| Result :     | Pass   |